Week 5

① Normalizing Adjacency Matrix.

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \qquad D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\tilde{A} \to \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$\tilde{D} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\tilde{D}^{-\frac{1}{2}} = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & \frac{1}{\sqrt{3}} & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\hat{A} = \tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}}$$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & \frac{1}{\sqrt{3}} & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & \frac{1}{\sqrt{3}} & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\to \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & \frac{1}{\sqrt{3}} & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{2} & \frac{1}{\sqrt{6}} & 0 \\ \frac{1}{\sqrt{6}} & \frac{1}{3} & \frac{1}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{6}} & \frac{1}{2} \end{bmatrix}$$

$$\hat{A} = \begin{bmatrix} \frac{1}{2} & \frac{1}{\sqrt{6}} & 0 \\ \frac{1}{\sqrt{6}} & \frac{1}{3} & \frac{1}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{6}} & \frac{1}{2} \end{bmatrix}$$

- Since we see the factor of degree goes to denominator, means the terms Relate to high degree node becomes smaller.

A node with many degree might complete dominates its neighbour feature & during aggregation.
So scaling its effect down, helps us gauge about other nodes.

Q3) we know, eigen value of ~~M~~ is $0 < \lambda_M < 2$

$$L = I_N - \underbrace{D^{-1/2} A D^{-1/2}}_{M}.$$

$$= I_N - M \quad \cancel{\text{is } 0 < \lambda_M < 2}$$

$$\therefore Mv = \lambda_M v$$

$$Lv = \lambda_M v$$

$$(I_N - M)v = I_N v - Mv = \cancel{I_N - \lambda_M}\, v$$

$$= \cancel{(I_N - M)v}$$

$$= (I_N - \lambda_M)v$$

~~\therefore~~ v is also eigen vector of L.

$$\lambda_L = 1 - \lambda_M$$

$$0 \leq \lambda_L \leq 2.$$

$$\boxed{-1 \leq \lambda_M \leq 1}$$

→ If we dont add I to A, we know after multiplying X, it ~~self~~ aggregates stuff but neighbours.
It would its own information.